

Orbital Mechanics Golf

Demo 7 of 16 · Part four, Example projects · 20 minutes for stages 1–3, plus the Sandbox

You need: `index.html` open in any modern browser — projector or phone. No network, no install, no accounts; nothing is fetched at run time and nothing leaves the machine. Chromebook-friendly.

Learning goals

One per stage, in the words the learner reads on the Takeaway card. Nothing else to state and nothing to read out: each is what its stage demonstrates, and each stage's check-yourself item tests exactly that.

- **Burn at apoapsis.** A prograde burn changes the orbit half a revolution away: burn at apoapsis and periapsis rises, burn at periapsis and apoapsis rises.
- **Escape at periapsis.** The same Δv buys energy in proportion to how fast you already are, so escape is cheapest at periapsis: 661 m/s here against 1,835 m/s from apoapsis. That is the Oberth effect.
- **Finite thrust.** Spread over time, a burn is cheapest centred on periapsis — 775 m/s centred against 919 m/s lit at periapsis — and with a weak engine six short kicks cost 731 m/s where one long burn costs 1,498.
- **Sandbox.** Every strategy that works, works for one reason: it spends Δv where v is already large.

Where this sits on the path

The first of Part four's ten AI-built classroom tools, and the first of its two games. It carries no handover from Demo 6: what Part four shares is not a chain of ideas but a discipline — each tool starts from a learning goal, exposes its assumptions, and makes its evidence available for challenge. This one's engine is a real gravity integrator, which is why its central result can be measured rather than asserted.

One question for the session

Ask it before anyone touches a control; return to it at the debrief.

"Escaping Earth needs a fixed amount of energy. You have a fixed tank. So does it matter where in the orbit you spend it?"

Most of the room says no, and the arithmetic they have — Δv is Δv — supports them. Stage 2 is them finding out the question was about energy, not speed.

Running it

Open  **Settings** → **Presentation mode** first: it scales the page for the back row and adds a  **Notes** button. The **Guide** card opens on load and explains the controls.

- **0–3 min** Stage 1 together on the projector. Take the room's prediction by show of hands first. Fly Level 1: → **Ap**,  **Plan burn**, drag Δv , read the dotted preview, commit. Vocabulary as you go: periapsis (low, fast), apoapsis (high, slow), prograde.
- **3–8 min** Students fly Levels 1–2 in pairs, watching the **Physics HUD**'s bottom row — "energy gained per 1 Δv " — as the ship falls.
- **8–16 min** The main event: stage 2, **Level 3 "Escape Artist"**. The stage withholds the flight view until a prediction is committed; say so out loud. Par is 791 m/s and the tank holds 2,057, so the wasteful route still works — it just costs. When someone makes par, have them say *where* they burned.
- **16–19 min** Debrief from a student's energy-vs-time plot and the **show the math** panel under it, which evaluates $\Delta \epsilon = v \cdot \Delta v + \frac{1}{2} \Delta v^2$ on their own numbers and checks it against the simulation. The burn is a vertical jump in ϵ , taller when v

was larger for the *same* Δv .

- **19–20 min** Post the leaderboard. Stage 3 unlocks once a student has escaped; Levels 6–7 make a second session or homework.

Stage by stage

Stage 1 — Burn at apoapsis (Levels 1–2)

Question

You are at the top of an ellipse and you push forward. Which part of the orbit moves?

Predict prompt

Burn forward at apoapsis. Does periapsis rise, does apoapsis rise, or do both? Three buttons, captured, echoed beside the result.

Activity

Level 1 circularizes an ellipse; Level 2 raises apoapsis into a band. Together they show the rule in both directions.

Expected observation

The dotted preview lifts the *far* side while the near side barely moves. The observation cue fires at each apsis with the speed there, naming apoapsis as the minimum energy-per- Δv point.

Check-yourself answer key

Because it raises the point half an orbit away, which from apoapsis is periapsis. The two wrong options are "apoapsis is where the engine is most efficient" (the refutation item; the opposite is true and stage 2 measures it) and "a single burn reshapes the whole orbit at once".

Teaching limit

Do not let this stage become the lesson. It is vocabulary and geometry; nothing about cost is established yet, and a student who leaves thinking apoapsis is "the good place to burn" has acquired the misconception stage 2 exists to break.

Stage 2 — Escape at periapsis (Levels 3–5)

Question

Escaping Earth needs a fixed amount of energy — so does it cost the same Δv wherever you burn?

Predict prompt

Level 3 starts on an ellipse with periapsis 11,100 km and apoapsis 70,100 km above the surface. Which burn point escapes for less Δv ? **The flight view does not appear until this is answered.**

Activity

→ **Pe**, 🔥 **Plan burn**, push Δv until the **ORBIT** chip flips from BOUND to ESCAPING. Then have them undo and try it from apoapsis on the same tank: that comparison is the stage. Levels 4 and 5 apply the same idea to a transfer and a powered flyby.

Expected observation

661 m/s from periapsis, 1,835 from apoapsis, a factor of 2.77 — both minima found against this engine. The HUD's "energy gained per 1 Δv " reads 6.08 km/s at periapsis and 1.39 at apoapsis: the same numbers, live.

Check-yourself answer key

Item 1: **because you are moving fastest there, and the energy added is $v \cdot \Delta v + \frac{1}{2} \Delta v^2$.** The refutation option is "because you are closer to Earth" — see Misconception 1. Item 2: **no — an unpowered assist borrows the moon's motion; Oberth is about when you spend fuel.** See Misconception 2.

Teaching limit

$2.77\times$ belongs to *this* ellipse. Round the orbit out and the gap shrinks; on a circle there is no periapsis to prefer. What generalises is the direction of the effect, not its size, and the Details drawer says so in as many words.

Stage 3 — Finite thrust (Levels 6–7)

Question

Your engine needs hours to deliver its Δv . Where in the orbit should that burn happen?

Predict prompt

The full burn lasts a noticeable arc. Light it exactly at periapsis, or start before periapsis?

Activity

Levels 1–5 use impulsive burns; 6 and 7 give the engine 0.055 and 0.013 m/s². Three readouts let students compute an ignition point instead of guessing: the HUD's **Time to periapsis**, the planner's **Burn duration**, and the **lead** box beside → **Pe**, which stops the warp that many minutes short. Locked until a student has escaped in stage 2.

Expected observation

Level 6: 775 m/s centred on periapsis, 919 lit at periapsis, against an impulsive ideal of 661 — finite thrust always costs something and timing decides how much. Level 7's engine is weak enough that one long burn smears round most of the orbit for 1,498 m/s, where six kicks of roughly 2.8 hours each cost 731.

Check-yourself answer key

Because all of it happens after the fast point, at falling speed. The refutation option is "a longer burn wastes more fuel as heat" — see Misconception 4.

Teaching limit

Steering losses are ignored entirely, and there is no rocket equation, so the finite-thrust penalty here has the right shape and not the right size. Treat 775-against-919 as "centring matters, and roughly this much".

Stage 4 — Sandbox (outside the path)

No goal, no par, 200 units of Δv , all three engines selectable, a second fictional moon, nothing scored. Its Predict card asks for a number rather than a button — how cheaply do you think you can escape? — and its check-yourself item is the synthesis one: which single quantity decided every answer across the three stages. The answer is your speed at the instant you spend the Δv .

The physics

An impulsive burn changes speed by Δv along the velocity, so kinetic energy per unit mass goes from $\frac{1}{2}v^2$ to $\frac{1}{2}(v+\Delta v)^2$:

$$\Delta \epsilon = \mathbf{v} \cdot \Delta \mathbf{v} + \frac{1}{2}\Delta v^2$$

The $\mathbf{v} \cdot \Delta \mathbf{v}$ term dominates, and it carries your speed at the instant of the burn. Gravitational potential is unchanged at that instant — you have not moved — so the whole difference between a cheap burn and an expensive one is the v in that term. On an ellipse, speed peaks at periapsis.

The HUD's "energy gained per 1 Δv " readout is therefore numerically just your current speed — worth asking a calculus class to differentiate $\frac{1}{2}v^2$ and see why.

Where does the extra energy come from? The propellant's chemical energy is fixed; what changes is how it divides between ship and exhaust. At high speed the engine acts over more distance per second, so more of it ends up in the ship and less in the plume.

Misconceptions to catch

Misconception 1: "The burn is better at periapsis because you are *closer* to the planet." Proximity is not the mechanism — **speed** is. $\Delta \epsilon = \mathbf{v} \cdot \Delta \mathbf{v} + \frac{1}{2}\Delta v^2$ contains v , not r . Being deep in the well matters only because falling there is what made the ship fast; a ship equally fast far from the planet gains just as much from the same burn. It is the wrong option on stage 2's first check item.

Misconception 2: "The Oberth effect is the same as a gravity assist." An **unpowered assist** (Level 5's moon) takes energy from the *moon's orbital motion* — no fuel, and the moon slows imperceptibly in exchange. An **Oberth burn** spends fuel and merely spends it at the most profitable moment. A powered flyby combines both, which is what Level 5 actually is.

Misconception 3: " Δv is like distance — it costs the same anywhere." The Δv *spent* is the same anywhere; the *energy it buys* is not. That is the whole game, and it is the third wrong option in stage 2.

Misconception 4: "A long burn costs more because the engine wastes fuel." The engine never changes within a level. What changes is the speed at which each second of thrust is delivered. It is stage 3's refutation item.

Misconception 5: "A very tall ellipse has escaped." Escape is $\epsilon > 0$, not a distance. A student who crosses the system edge while still bound is told so on screen, with their actual ϵ quoted, because the geometry is genuinely misleading.

Discussion questions

- Level 3's energy plot shows a taller jump in ϵ for the same Δv at periapsis. Which term of $\Delta \epsilon = \mathbf{v} \cdot \Delta \mathbf{v} + \frac{1}{2}\Delta v^2$ explains this, and what would have to be true for the other term to matter more?
- The HUD says "energy gained per 1 Δv = your current speed". Differentiate $\frac{1}{2}v^2$ and show why.
- Why does a prograde burn at periapsis raise the *apoapsis*, on the far side of the orbit?
- In Level 5, where does the escape energy come from when you fly close behind the moon without burning? Name the thing that slows down in exchange.
- A mission that wants to be captured into orbit has to *lose* energy. Where on the approach should it burn, and does the same argument run backwards? (SMART-1: perigee thrust arcs outbound, braking around perilune inbound.)

6. Could you escape with a radial-out burn? Try it, then explain from the formula why pushing "straight up" wastes fuel.

Numbers, and where each one comes from

Every figure in the demo carries one of these labels on its face.

Quantity	Value	Provenance
Earth μ , radius	398,600 km ³ /s ² · 6,371 km	Sourced. NASA NSSDC Earth Fact Sheet, read 2026-09-16: "GM ($\times 10^6$ km ³ /s ²) 0.39860", "Volumetric mean radius (km) 6371.000".
Level pars	L1 712 · L2 1,107 · L3 791 · L4 1,819 · L5 1,424 · L6 838 · L7 823 m/s	Measured against this exact engine. The integrator has no random element, so there is no seed: re-running test-physics.js reproduces every figure.
L3 escape minima	661 m/s at periapsis · 1,835 at apoapsis	Measured by bisection; the periapsis figure also matches closed-form $v_{\text{esc}} - v_p$ to three figures.
L6 / L7 finite thrust	775 centred · 919 lit at Pe · 731 in six kicks · 1,498 in one burn	Measured , same harness.
The moon	invented orbit, mass and size	Illustrative. Not our Moon — closer and heavier relative to this setup, so an assist fits a class period. Labelled on the map itself.
Low-thrust stages escape by repeated perigee burns	—	Sourced (the practice): ESA's SMART-1 page, 31 Aug 2006, read 2026-09-16 — "thrust arcs around the perigee and coast arcs around the apogee". The motive is Reasoned from 731-against-1,498 and MIT OCW 16.07 L17 (read 2026-09-16).

Do not quote these numbers as spaceflight figures. The model has no atmosphere, no mass loss and so no rocket equation, no steering cost, and no third-body perturbation beyond the one fictional moon. It isolates one relationship by stripping away everything that would compete with it.

Presenter controls

- **⚙ Settings** holds the three presenter controls in order — **Open Presenter Notes** (this page, on screen), **Presentation mode** (larger type plus a header Notes button) and **Reset**.
- **Reset** restores a first load in place, with no page reload: Level 1 on the pad, an empty Δv tally, trajectory and undo stack cleared, warp back to 1 $\times$, the HUD open, all three predictions forgotten, stage 3 locked again, badges and saved name cleared. It leaves two things alone on purpose: **presentation mode**, since re-shrinking the projector between rooms would be hostile, and the saved **class leaderboards**, which hold other students' entries — use *Clear scores*, which asks first.
- **Keys:** Space pause · B burn planner · Enter commit · Esc cancel · Z undo · X cut engine · , · . warp down/up · P / A warp to Pe / Ap · H HUD · F follow camera · M briefing · R restart level · ? Guide · N Presenter Notes.
- **Undo** rewinds the last burn and refunds the fuel, and is offered again after a crash. Students should iterate, not restart.
- The four stage tabs jump between the lesson's beats; picking a level from **🚀 Levels** moves the tabs with it. A #stage-3 link clamps to the furthest unlocked stage rather than routing around the gate.

